



7.1.6 Wrap-Up: Ticket Out the Door

1. Write three equivalent expressions for $12a^3b^5 - 16a^2b^2 + 20a^3b$. One of the expressions should include the greatest common monomial factor.

Unit 7, Lesson 2: Factoring a Trinomial with Algebra Tiles



Benchmark

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.



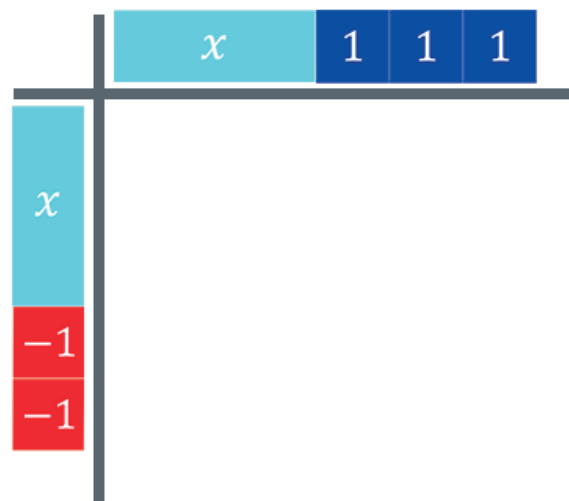
Learning Target

- ☐ I can use algebra tiles to factor a trinomial with a leading coefficient of 1.



7.2.1 Warm-Up: Multiplying Binomials

1. An algebra tile mat is shown with two binomial factors.

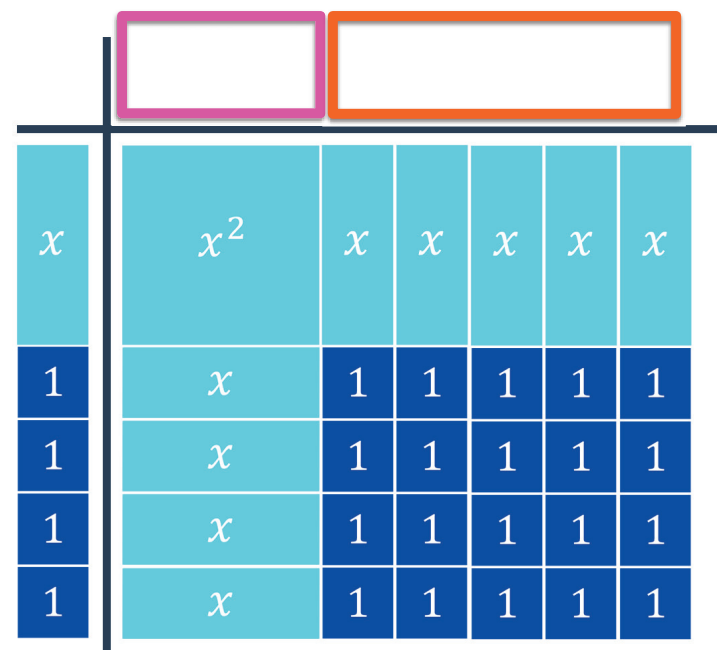


- What are the two binomials represented?
- What is the product of the two binomials?



7.2.2 Exploration: Model Factoring with Algebra Tiles

1. An algebra tile mat is shown, where the product is the trinomial $x^2 + 9x + 20$ and one of its factors is $(x + 4)$.



Diego and Maria Jose both justified how they determined what belonged in the **pink** box in the following ways.

Diego	Maria Jose
I found $\frac{x^2}{x}$ because x^2 is the product of x and something else.	I asked myself what times x would give me x^2 .

a. Discuss with your partner both justifications.

b. Use one of the methods to complete the binomial for the top bar of the algebra tile mat.

	Drawing	Algebraic Representation
Pink Box		x
Orange Box		
Binomial Factor		

2. Discuss with your partner how to determine the two binomial factors when only given the product. Summarize your discussion.

x^2	x	x	x	x	x
x	1	1	1	1	1
x	1	1	1	1	1
x	1	1	1	1	1
x	1	1	1	1	1

The two binomials are the factors of the trinomial $x^2 + 9x + 20$. The process of finding the two binomials is called **factoring**.

3. The algebra tiles shown represent the trinomial $x^2 + 9x + 18$.

x^2	x	x	x
x	1	1	1
x	1	1	1
x	1	1	1
x	1	1	1
x	1	1	1
x	1	1	1

- a. Discuss with your partner what the similarities and differences are between the algebra tiles that represent $x^2 + 9x + 20$ and $x^2 + 9x + 18$. Summarize your discussion.

Similarities	
Differences	

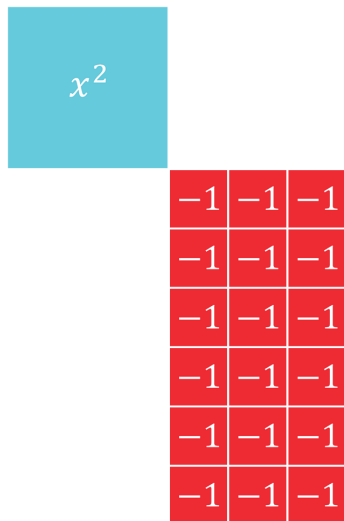
- b. What are the two binomials that are factors of $x^2 + 9x + 18$?

4. The table shows the algebra tile representation for two trinomials. Work with your partner to determine the two binomial factors.

Trinomial	$x^2 - 9x + 18$	$x^2 - 3x - 18$																																																								
Algebra Tiles	<table><tr><td>x^2</td><td>$-x$</td><td>$-x$</td><td>$-x$</td></tr><tr><td>$-x$</td><td>1</td><td>1</td><td>1</td></tr><tr><td>$-x$</td><td>1</td><td>1</td><td>1</td></tr><tr><td>$-x$</td><td>1</td><td>1</td><td>1</td></tr><tr><td>$-x$</td><td>1</td><td>1</td><td>1</td></tr><tr><td>$-x$</td><td>1</td><td>1</td><td>1</td></tr><tr><td>$-x$</td><td>1</td><td>1</td><td>1</td></tr></table>	x^2	$-x$	$-x$	$-x$	$-x$	1	1	1	$-x$	1	1	1	$-x$	1	1	1	$-x$	1	1	1	$-x$	1	1	1	$-x$	1	1	1	<table><tr><td>x^2</td><td>x</td><td>x</td><td>x</td></tr><tr><td>$-x$</td><td>-1</td><td>-1</td><td>-1</td></tr><tr><td>$-x$</td><td>-1</td><td>-1</td><td>-1</td></tr><tr><td>$-x$</td><td>-1</td><td>-1</td><td>-1</td></tr><tr><td>$-x$</td><td>-1</td><td>-1</td><td>-1</td></tr><tr><td>$-x$</td><td>-1</td><td>-1</td><td>-1</td></tr><tr><td>$-x$</td><td>-1</td><td>-1</td><td>-1</td></tr></table>	x^2	x	x	x	$-x$	-1	-1	-1	$-x$	-1	-1	-1	$-x$	-1	-1	-1	$-x$	-1	-1	-1	$-x$	-1	-1	-1	$-x$	-1	-1	-1
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Binomial Factors																																																										

One common attribute of all of the algebra tile representations is that they are all rectangles.

5. Consider the following algebra tile representation.



- Complete the algebra tile representation of a trinomial by drawing and labeling x or $-x$ tiles.
- What is the trinomial that is represented by the completed drawing?
- What are the binomial factors of the trinomial?

6. The algebra tile representation of two different trinomials is shown.

a. Complete the table.

Algebra Tiles	Trinomial	Factors

Each of the trinomials in the table has either twenty-four unit squares, $+1$, or twenty-four negative unit squares, -1 . Also, each trinomial has some combination of four and six $+x$ or $-x$ tiles.

- b. Discuss with your partner how the x -term of each trinomial relates to the four and six ^+x or ^-x tiles.
- c. Discuss with your partner how the constant term, 24 or -24 , of each trinomial relates to the four and six ^+x or ^-x tiles.

7. Consider the trinomial $x^2 - 5x - 24$.

- a. Work with your partner to determine how many ^+x or ^-x tiles are needed for $x^2 - 5x - 24$.

^+x tiles	^-x tiles

- b. Draw a diagram of the algebra tiles that show the trinomial $x^2 - 5x - 24$.

- c. Use your algebra tiles diagram to choose the number of ^+x or ^-x tiles that match your diagram. Then, fill in each blank with the appropriate values.

	x	<u> </u>
x	x^2	☐ $1x$ ☐ $-1x$ ☐ $6x$ ☐ $-6x$ ☐ $2x$ ☐ $-2x$ ☐ $8x$ ☐ $-8x$ ☐ $3x$ ☐ $-3x$ ☐ $12x$ ☐ $-12x$ ☐ $4x$ ☐ $-4x$ ☐ $24x$ ☐ $-24x$
<u> </u>	☐ $1x$ ☐ $-1x$ ☐ $6x$ ☐ $-6x$ ☐ $2x$ ☐ $-2x$ ☐ $8x$ ☐ $-8x$ ☐ $3x$ ☐ $-3x$ ☐ $12x$ ☐ $-12x$ ☐ $4x$ ☐ $-4x$ ☐ $24x$ ☐ $-24x$	-24

- d. Discuss with your partner how using the factors of -24 , and the value of the coefficient of the b -term, -5 , is helpful when factoring the trinomial $x^2 - 5x - 24$. Summarize your discussion.



7.2.3 Guided Practice: Factoring Trinomials

1. Complete the table for $x^2 + 18x - 40$.

	x	_____
x	x^2	☐ $1x$ ☐ $-1x$ ☐ $8x$ ☐ $-8x$ ☐ $2x$ ☐ $-2x$ ☐ $10x$ ☐ $-10x$ ☐ $4x$ ☐ $-4x$ ☐ $20x$ ☐ $-20x$ ☐ $5x$ ☐ $-5x$ ☐ $40x$ ☐ $-40x$
_____	☐ $1x$ ☐ $-1x$ ☐ $8x$ ☐ $-8x$ ☐ $2x$ ☐ $-2x$ ☐ $10x$ ☐ $-10x$ ☐ $4x$ ☐ $-4x$ ☐ $20x$ ☐ $-20x$ ☐ $5x$ ☐ $-5x$ ☐ $40x$ ☐ $-40x$	-40

2. Complete the table for $x^2 + 12x + 32$.

	x	_____
x	x^2	☐ $1x$ ☐ $-1x$ ☐ $8x$ ☐ $-8x$ ☐ $2x$ ☐ $-2x$ ☐ $16x$ ☐ $-16x$ ☐ $4x$ ☐ $-4x$ ☐ $32x$ ☐ $-32x$
_____	☐ $1x$ ☐ $-1x$ ☐ $8x$ ☐ $-8x$ ☐ $2x$ ☐ $-2x$ ☐ $16x$ ☐ $-16x$ ☐ $4x$ ☐ $-4x$ ☐ $32x$ ☐ $-32x$	32



7.2.4 Your Turn!

1. Complete the table for $x^2 - 11x + 18$.

	x	_____
x	x^2	☐ $1x$ ☐ $-1x$ ☐ $6x$ ☐ $-6x$ ☐ $2x$ ☐ $-2x$ ☐ $9x$ ☐ $-9x$ ☐ $3x$ ☐ $-3x$ ☐ $18x$ ☐ $-18x$
_____	☐ $1x$ ☐ $-1x$ ☐ $6x$ ☐ $-6x$ ☐ $2x$ ☐ $-2x$ ☐ $9x$ ☐ $-9x$ ☐ $3x$ ☐ $-3x$ ☐ $18x$ ☐ $-18x$	18



7.2.5 Wrap-Up: Reflection

1. How “in the zone” do you feel right now about factoring trinomials?



Unit 7, Lesson 3: Factoring a Trinomial with a Leading Coefficient of 1



Benchmark

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.



Learning Target

- ☐ I can factor a trinomial with a leading coefficient of 1.